Determining the Size and Stiffness of Objects Using Underactuated Grasping

Matthew W. Hunter
Georgia Institute of Technology

Introduction

Motivation

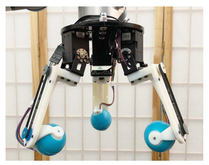
- Machines are implemented to perform tasks humans cannot
- Systems that replicate how humans interact with the environment are less common
- Through haptic feedback touch sensation can be transferred from the environment, through a machine and into humans
- **Technology has implementation in construction, surgery, the military and prosthetics**

Background

Fully Actuated vs Under Actuated

- All degrees of freedom controlled by an actuator
- Tend to be heavier
- Increased cost

- Movement through soft joints and wires
- Lighter
- Less actuators
- Puppet-like



Force Sensitive Resistor (FSR)

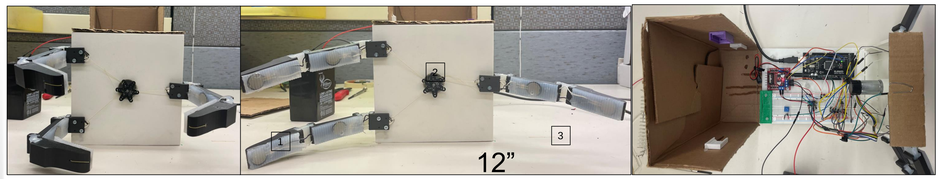
- Similar to a potentiometer
- Pressure changes the resistance of the component with voltage proportional to pressure
- Good for measuring relative pressure instead of exact unit measurements

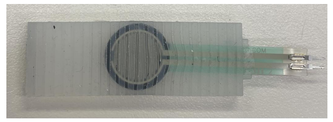
Haptics

- Use of vibration motors to mimic sensation
- VR companies use haptic technology to allow users to interact with virtual environments
- Video game controllers use haptic feedback to create a more engaging experience through vibration

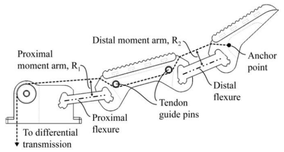
Methods

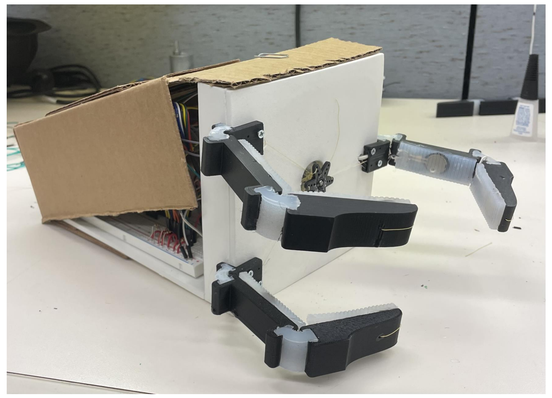
Design





1. FSR under a silicone grip pad



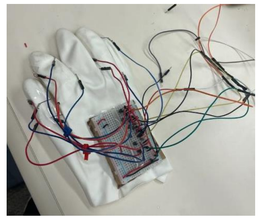


2. Actuation wire diagram

3. 3D printed finger with soft silicone joints

Potential haptic device

- vibration motors match FSR placement
- More pressure on FSR equals more vibration

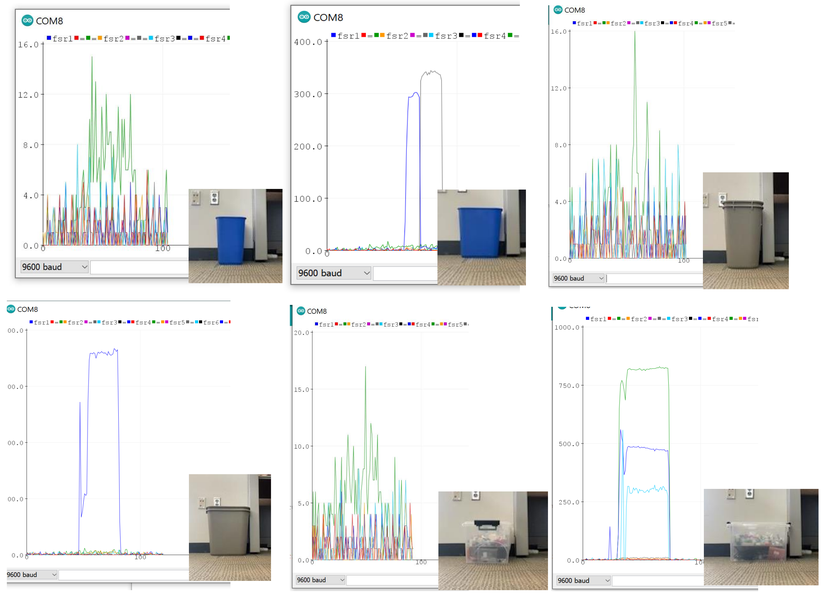


Experiment

- Used three rounded prisms
- Each has a long and short side
- Each has a different stiffness
- Six sizes, three Stiffnesses

Conclusion

Results



- Small Objects: At rest readings but scaled up
- Large Objects : Reading spikes and stays stable
- Stiff Objects: Much higher FSR Values
- Compliant Objects: Lower FSR Values

FSR 1
FSR 2
FSR 3
FSR 4
FSR 5
FSR 6

Conclusion

- FSRs integrated to an under actuated hand
- Differences between size stiffness recorded
- Proof of concept for haptic feedback

Future Works

- Implement additional and improved sensing
  - -temperature
  - -capacitance
  - - bend
- Additional shapes
- Human trial haptic testing to determine how effective system is in recreating sensation

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